



Cornell University
College of Agriculture
and Life Sciences

Department of Natural Resources
Cornell Biological Field Station
900 Shackelton Point Rd.
Bridgeport, New York 13030-9791
t. 315.633.9243
f. 315.633.2358
www.dnr.cornell.edu/fieldst/cbfs.htm

Anne Scofield, Biology Technical Lead
Nicole Singleton, Project Officer
Elizabeth Hinchey
US EPA, Great Lakes National Program Office (G-17J)
77 W. Jackson Blvd., Chicago, IL 60604

September 26, 2026

RE: Zooplankton Data (CSMI 2024 Erie Nearshore Spring and Summer)
Grant **GL-00E03089**

Dear Anne, Nicole, and Beth,

We are pleased to submit the D20 zooplankton data for the 2024 Spring and Summer CSMI Lake Erie Nearshore samples. These were 12 nearshore sites collected for both spring and summer in Lake Erie for the CSMI Year 2024. Please find attached the data file in Excel spreadsheet format ZOOP_DATA_2024_NS Erie_260926. Please find below all supporting information that starts with this cover letter.

The format of this data submission includes tabs for zoosam (sample information), zoocnt (density and biomass for each taxa, length measurements of individuals), and zoofact (species list with biomass equation coefficients, October 2, 2024 matching GLNPO Zooplankton Database 7.0). Not included are typical tables for Default Meas, and QA (Percent Similarity of Community (PSc) values for QA samples. All fields are defined in adjoining worksheets.

Field operations were conducted following EPA SOP LG 402A (Revision 15, October 2020).

- Sampling in 2024 on the R/V Lake Guardian occurred on Lake Erie April 17-20 and August 10-14. In Spring, 11 of the 12 nearshore sites were identified with the prefix ECSER ending with lower case n. One site was ER95b which is a routine summer

benthic site. In summer, for eleven of the twelve nearshore sites the prefix ECS was used and ER95b was the twelfth sample. For nearshore samples, a single 63 micron net was collected for the whole water column 2 m off the bottom to the surface.

- Flowmeter performance was closely monitored during the surveys. Flowmeters for D20 and D100 nets were calibrated together on a single empty net ring at the beginning of the survey with twenty 20 m tows. A check of this calibration was done with 10 tows at the midpoint of each survey, and the value should be within 10% of the initial calibration.

TSK (Tsurumi-Seiki Co., Ltd.) flowmeters were used for the 2024 surveys. For D20 nets, Flowmeter Unit #7207 was used for Spring and Unit #7294 was used for Summer. In Spring, a flowmeter calibration for #7207 was done on April 10, 2024 at Lake Michigan site MI11, yielding an average value of 130.1 counts (5.0 s.d.) or a FMCALIB of 0.154. The midcruise check yielded 139.6 counts (11.8 s.d.). For Summer, an initial flowmeter calibration was done on August 1, 2024 for unit #7294, yielding an average value of 134.3 counts (s.d. 12.9) or a FMCALIB of 0.149. The midcruise check was conducted at SU01 on September 23, 2024 with a calibration of 134.2 counts (s.d. 6, n=5). Both seasons were within the 10% difference from the initial calibration required. Net efficiency was lower as expected for the finer D20 net than the D100 net, 72% for Spring and 69% for Summer for all lakes.

Laboratory analysis for 2024 D20 samples followed the modified CSMI sample protocol from the typical LG403 SOP. All individuals of large zooplankton (*Mysis*, *Bythotrephes*, *Cercopagis*) were removed and counted as for LG403. The aliquot subsampling method was used for subsampling a known volume with a wide bore pipette for crustacean species which is identified in SPLITA in zoosam. The total volume and aliquot volume used to calculate SPLITA for each sample are listed in the tab 'zoop dilution info'. The aliquot method was also used for rotifers, veligers, and nauplii as in LG403 using a different subsample and are listed in zoosam. Therefore we include data for both crustaceans and microzooplankton rotifers, veligers, and nauplii.

-Default values for individual dry weight biomass were not needed.

-The nonnative predatory cladoceran *Cercopagis pengoi* were abundant (>500) in one summer nearshore summer Lake Erie sample (ECS932n). *Bythotrephes longimanus* was abundant (>200) in one sample ECS951n. All individuals in every sample were counted without subsampling.

-The SPECCODE (DIPFLUV) was used for detected species *Diaphanosoma fluviatile* (Whitmore et al. 2019)- for 2024 found at five nearshore sites in Lake Erie. Other individuals at this site and others in Lake Erie were identified as DIPBIRG or DIPSP.

Electronic zooplankton analysis files for these nearshore Spring and Summer 2024 samples were provided in the portal as the compressed folders "Erie Spring 2024 NS" and "Erie Summer 2024 NS".

Please let me know if you have any questions about this submission.

Sincerely,
James Watkins
Cornell University